

iHP *Iterate to shine brighter.*

No man is an island, entire of itself; every man is a piece of the continent, a part of the main. —John Donne

Stakeholders

- Identification & classification

The problem our team aims to solve lies within people, so the design inevitably revolves around people. Therefore, it is imperative to conduct a detailed and comprehensive analysis of the stakeholders that the project can reach, and may effect us. Only through continuous communication, reflection with people in the real world and improvement can we be able to confirm that our design is "responsible and good for the world".

Based on the core issues identified in the project and the preliminary design, we have gone out of our way to think of all groups that may be connected to our project and those that may be reached through arranged activity. After identification, we have classified these stakeholders, which are organized as follows:

Category	Stakeholders & Roles
Patients	- SCLC patients (currently being treated & not in treatment & potential) - Patients' family members & caregivers - Patient advocacy groups for lung cancer
Healthcare	- Medical oncologists, pulmonologists, thoracic surgeons - Nurses - Pharmacists, clinical laboratory staff - Hospital & clinic administrators - Ethics committees (IRB/EC)

Academia	- Principal investigators (PIs), research professors - University biosafety & ethics experts - Academic collaborators in oncology, synthetic biology, bioinformatics
Industry	- Biotech & pharma partners (drug development & manufacturing) - CROs (Contract Research Organizations) for clinical trials - medical insurance company
Government	- Regulatory agencies - National cancer & health research funders - Public health authorities - Biosafety & biosecurity regulators
Other teams	- other iGEM teams - team members from previous iGEM teams
Public organizations	- Nonprofit cancer foundations & research charities - Public health organizations - Patient support associations - Science education & outreach organizations
Public	- General public - People who volunteer to help advocate for cancer patients or spread knowledge on media like online platforms - Members in aforementioned communities

Table. Stakeholder Classification

- Analysis

To better engage with stakeholders from different groups, we need to first understand their needs and their connections to the project, and then arrange subsequent activities based on this. We use the tool of Mendelow's Matrix to understand the relationships between different stakeholder groups and our project. It also guides our attitudes towards groups in different quadrants:

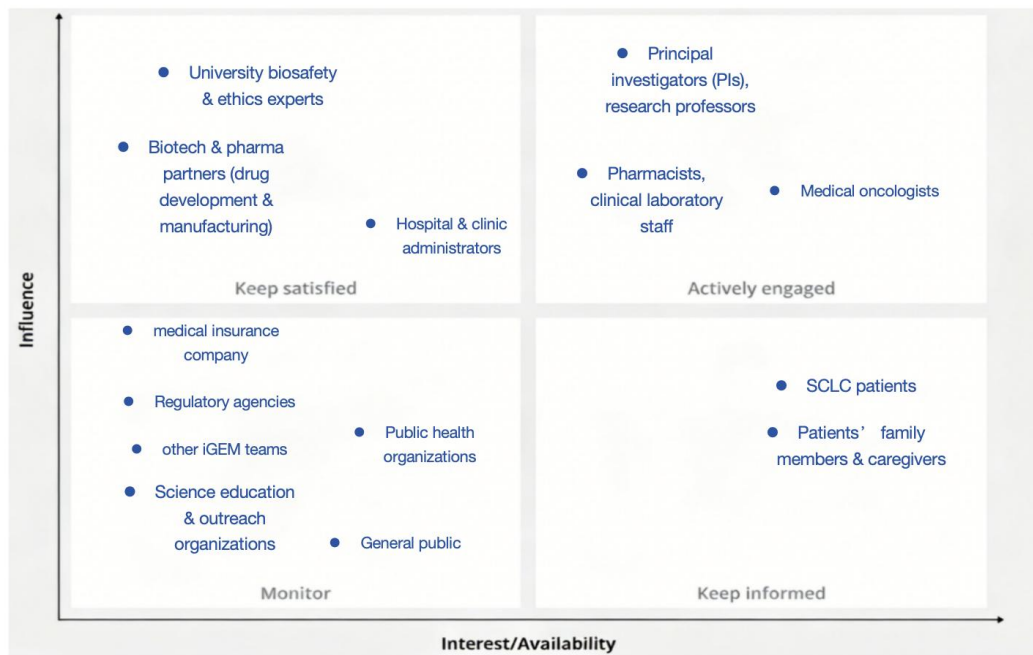


Figure 2 Mendelow's Matrix (showing how our stakeholders are related to the project)

After that, we specifically analyzed the characteristics of each stakeholder group to better understand "how our project will influence them, and how will they influence our project" .

Stakeholder Category	What They Value	What we want
Patients	Effective & safe treatment Improved survival Accessibility Affordability	-Their needs and expectations for a new therapy. -Their feedback on our prototype.

	Quality of life (Reduced side effects and convenience) Clear information Hope Confidentiality	-Their perception of cancer. How did they develop it and what would they recommend to those in a similar situation. -What other needs do they have, like mental support.
Healthcare	Clinical efficacy Safety Evidence-based data Ease of use Cost-effectiveness Integration into standard care	- What aspects should our project achieve to make it truly feasible in clinical context. -Their feedback on our prototype.
Academia	Scientific novelty Rigorous data Reproducibility	-Their feedback on our prototype. -Their advice on our design and experiments.
Industry	Commercial viability Scalability Intellectual property (IP) Market potential	-Their feedback on our prototype. -Their advice on our design and business plan.
Government	Safety Regulatory compliance Ethics	-Their advice on our design and safety module.

Other teams	Shared knowledge Collaboration opportunities Complementary technologies	-Their advice on our project. -The potential of cooperation.
Public	Transparency Responsibility Engagement	-Their perception on cancer. -Their perception on synthetic biology and its derived product .
Public organizations	Accessibility Advocacy alignment	-The current state of cancer patients. -The experience on public education. -The potential to collaborate.

- Project development timeline

With the inspiration from Team 2025 AFCM-egypt , we believe that integrating engagement with different stakeholders into our project development plan based on the needs of different groups can make our work more organized:

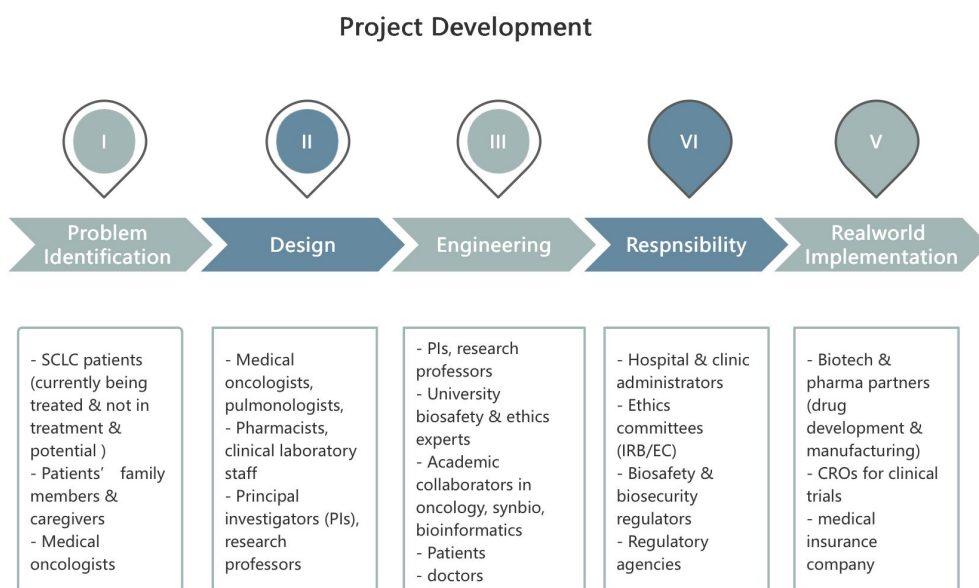


Figure 3 Project development and HP planning

iHP cycle

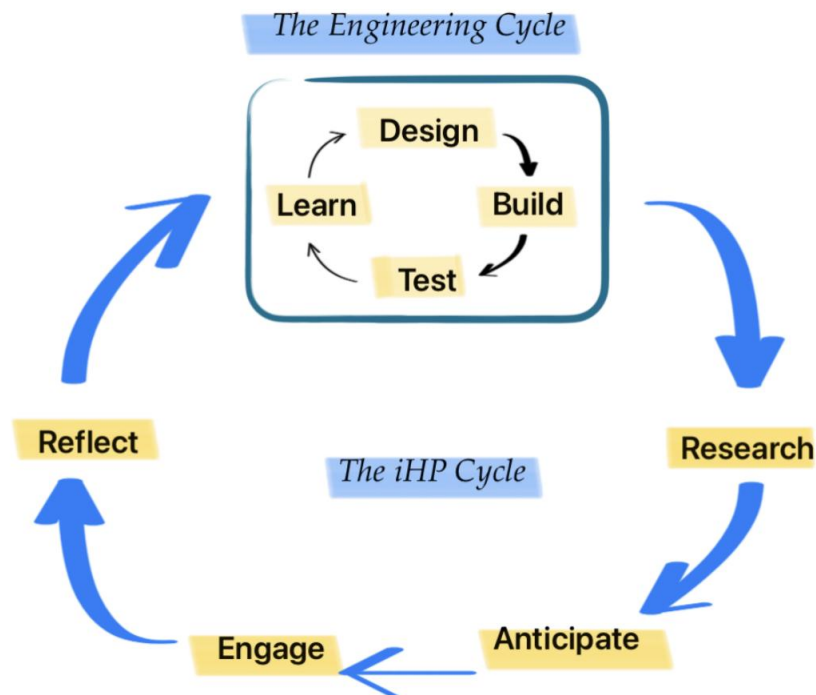


Figure 4. Iteration Diagram

By establishing a specific framework for iHP work, we hope that the implementation of relevant work will be carried out in an orderly and standardized manner.

Research: Conduct preliminary research based on the problems and requirements encountered during the implementation of projects such as engineering, analyze the stakeholder groups, etc., aiming to gain an understanding and judgment of the problems to be solved;

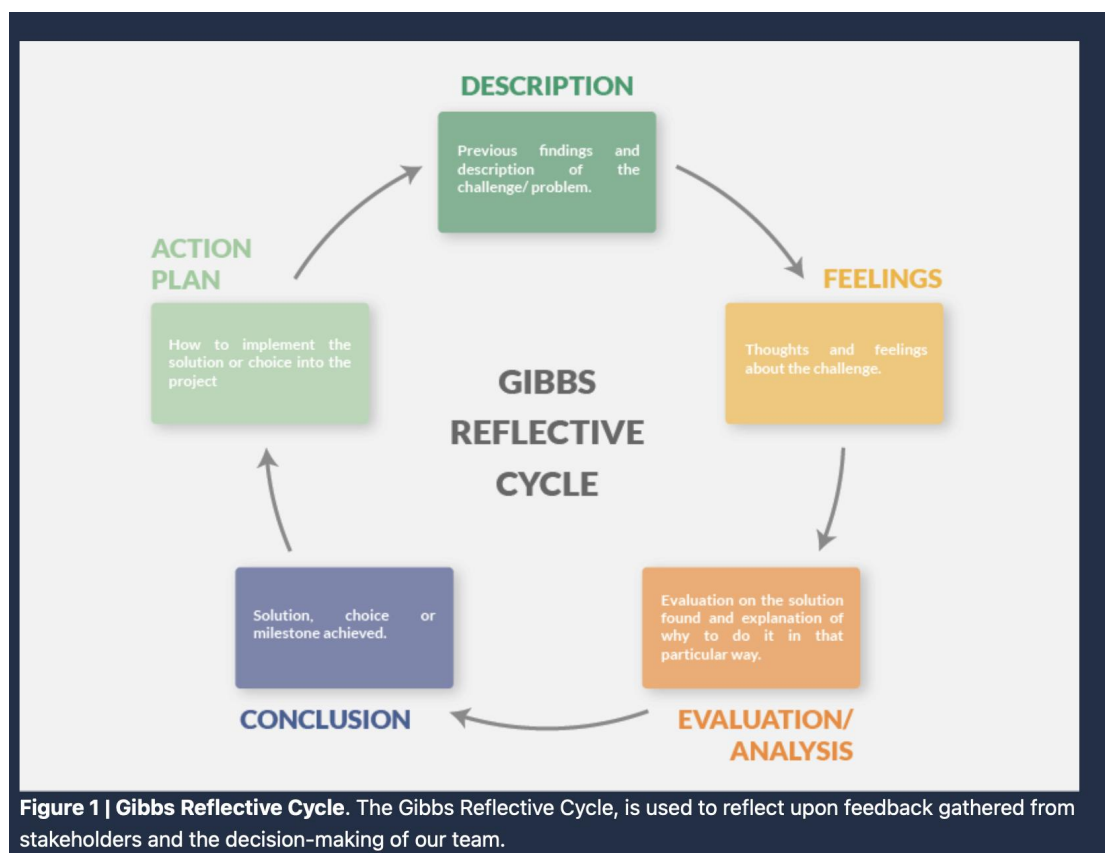
Anticipate: Based on the analysis of stakeholders and research results, identify at the current stage which stakeholders can provide important opinions and feedback for our project, their needs, through what channels we should communicate with them, and how to arrange things to better enable mutual assistance between the two parties, and then plan activities based on analysis and prediction;

Engage: Organize activities and conduct communication;

Reflect: Reflecting on the inspiration brought by the activities using a modular recording format helps us better clarify our thinking, guide the progress of the next step of work, and at the same time make the presentation of this part of the work clearer and more intuitive. The key takeaways from the Reflect section should ideally be reflected in our design iterations.

Reflection & documentation

We build upon the Gibbs Reflective Cycle used by the 2022 TU Eindhoven team. It presents a thinking process for activity reflection .



"Description - feelings - evaluation - conclusion - action plan" constitute the entire Reflection cycle, and the listed components are also key points that should be paying attention to when recording activities.

When organizing the work content at the end , records can be made based on the subjects (corresponding to the aspect of the problem to be solved/consulted), and multiple interactions with the same stakeholder can be together recorded under the same section (since they often belong to the same Gibbs Reflective Cycle), which can help demonstrate how our project achieves "closing the loop" for this group/issue.

iHP Records (Activity Plans)

Hospital Survey:

A tumor hospital near the school was selected, interviews were conducted with SCLC patients and department doctors respectively, and a questionnaire was developed for a larger patient population. The questionnaire mainly includes an evaluation form for current treatment methods and the main areas of concern regarding new therapies, and also collects some information that can be used in subsequent education and popular science modules, such as the psychological and life changes brought about by cancer diagnosis to patients and their families, what issues patients wish they had known before getting cancer, whether there are still some areas of confusion in the current treatment that patients hope doctors or social organizations can provide information on, etc.

Consulting experts during engineering cycle

- **Medical workers:** What are your views on existing therapies? Are you willing to choose engineered bacteria therapy, what do you think is the possibility of such a method in clinical practice, what difficulties will be encountered during implementation, and how much do you think the development potential of this method is in the future?
- **Non-medical workers/the general public:** Do you have a smoking habit? Have you heard about small cell lung cancer and the link between smoking and this disease? What are your views on using engineered bacteria technology to treat diseases? How do you think we should popularize this technology and make people accept it?
- **Patients and their families**
 - Investigate patients' acceptance of pulmonary inhalation as a drug delivery method (Questionnaire)
 - Acceptable range for the duration and cost of drug resistance testing
 - Understand the core needs of patients: efficacy / safety / side effects...